

$$u = -3x^2y + ay^3 + 2x^2 + by^2$$

$$\frac{\partial u}{\partial x} = -6xy + 4x = \frac{\partial v}{\partial y}$$

$$v = -3xy^2 + 4xy + f(x)$$

$$\frac{\partial u}{\partial y} = -3x^2 + 3ay^2 + 2by \rightarrow$$

$$\frac{\partial v}{\partial y} = -6xy + 4x + f'(y)$$

$$\frac{\partial v}{\partial x} = -3y^2 + 4y + f'(x)$$

$$= 3x^2 - 3ay^2 - 2by$$

$$b = -2$$

$$a = 1$$

$$f'(x) = 3x^2$$

$$f(x) = x^3 + C \quad f(0) = 0$$

$$u = -3x^2y + y^3 + 2x^2 - 2y^2$$

$$v = -3xy^2 + 4xy + x^3$$

$$f(z) = 2z^2 + z^3i$$

$$f(z) = \frac{2z^2 - z + 5}{(z+1)(z-3)}$$

$$= \frac{2}{z-3} - \frac{1}{z+1}$$

$$= \frac{2}{z(1-\frac{3}{z})} - \frac{1}{z^2(1+\frac{1}{z})}$$

$$= \frac{2}{z} \sum_{n=0}^{\infty} (\frac{3}{z})^n - \frac{1}{z^2} \sum_{n=0}^{\infty} (-\frac{1}{z})^n$$

$$= \sum_{n=0}^{\infty} \frac{2 \cdot 3^n}{z^{n+1}} - \sum_{n=0}^{\infty} \frac{(-1)^n}{z^{n+2}}$$

$$= \sum_{n=0}^{\infty} \frac{2 \cdot 3^n}{z^{n+1}} - \sum_{n=0}^{\infty} \frac{(-1)^n}{z^{n+2}}$$

$$\oint_{|z|=2} \frac{1}{(z-3)(z^5-1)} dz$$

$$= -\text{Res}$$

$$= -2\pi i \text{Res} \left[\frac{1}{(z-3)(z^5-1)}, 3 \right]$$

$$+ 2\pi i \text{Res} \left[\frac{1}{(z-3)(z^5-1)}, \frac{1}{z^5}, 0 \right]$$

$$= \frac{-2\pi i}{3^5-1} = \frac{-\pi i}{121}$$

$$e^z + 1 = 0$$

$$z = (\pi + 2k\pi)i$$

$$-3\pi$$

$$\oint_C \frac{e^{\frac{1}{3}z}}{1+e^z} dz$$

$$= 2\pi i \text{Res} \left[\frac{e^{\frac{1}{3}z}}{1+e^z}, e^{\pi i} \right]$$

$$= -\pi i + \sqrt{3}\pi$$

$$\int_{-\infty}^{+\infty} \frac{e^{\frac{1}{3}x}}{1+e^x} dx$$

$$= \int_{-\infty}^{+\infty} \frac{e^{\frac{1}{3}z}}{1+e^z} dz$$

$$= 2\pi i \sum \left[\frac{e^{\frac{1}{3}z}}{1+e^z}, z_k \right]$$

$$= 0$$

$$\text{六. } \begin{cases} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, & x < -3, y > 1 \\ u|_{y=1} = f(x), & x \leq -3 \\ u|_{x=-3} = g(y), & y \geq 1 \end{cases}$$

基础解系: $\frac{1}{2\pi} \ln \frac{1}{r}$

$$G = \frac{1}{2\pi} \ln \frac{1}{r_{M_0}} - \frac{1}{2\pi} \ln \frac{1}{r} - \frac{1}{2\pi} \ln$$

$$G = \frac{1}{2\pi} \ln \frac{1}{r_{M_0}} - \frac{1}{2\pi} \ln \frac{1}{r_{M_1}} - \frac{1}{2\pi} \ln \frac{1}{r_{M_2}} + \frac{1}{2\pi} \ln \frac{1}{r_{M_3}}$$

$$u(x_0, y_0) = - \oint f(x_0) \frac{\partial G}{\partial n_0} ds_0$$

$$= - \oint_{-3}^{+\infty} f(x_0) \frac{\partial G}{\partial y_0} dx_0 + \int_1^{+\infty} g(y_0) \frac{\partial G}{\partial x_0} dy_0$$

七. $u_{tt} = 4u_{xx} + x$

$$u_{tt} = 4u_{xx}$$

$$\left(\frac{dx}{dt}\right)^2 = 4 \quad \frac{dx}{dt} = \pm 2$$

$$\xi = x = 2t + C_1, \quad \eta = -2t + C_2$$

$$\xi = x - 2t, \quad \eta = x + 2t$$

$$x = \frac{1}{2}(\xi + \eta)$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta}$$

$$\frac{\partial u}{\partial t} = -2 \frac{\partial u}{\partial \xi} + 2 \frac{\partial u}{\partial \eta}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial \xi^2} + 2 \frac{\partial^2 u}{\partial \xi \partial \eta} + \frac{\partial^2 u}{\partial \eta^2}$$

$$\frac{\partial^2 u}{\partial t^2} = 4 \frac{\partial^2 u}{\partial \xi \partial \eta} - 8 \frac{\partial^2 u}{\partial \xi \partial \eta} + 4 \frac{\partial^2 u}{\partial \eta^2}$$

$$-16 \frac{\partial^2 u}{\partial \xi \partial \eta} = \frac{1}{2}(\xi + \eta)$$

$$\frac{\partial^2 u}{\partial \xi \partial \eta} = -\frac{1}{2}\xi - \frac{1}{2}\eta$$

$$\frac{\partial u}{\partial \xi} = -\frac{1}{2}\xi\eta - \frac{1}{4}\eta^2 + f(\xi)$$

$$u = -\frac{1}{4}\xi\eta^2 - \frac{1}{8}\eta^3 + \int f(\xi) d\xi + g(\eta)$$

$$u = f_1(\xi) + f_2(\eta) - \frac{1}{4}\xi\eta^2 - \frac{1}{8}\eta^3$$

$$= f_1(x-2t) + f_2(x+2t)$$

$$- \frac{1}{4}(x-2t)(x+2t)^2 - \frac{1}{8}(x+2t)^3$$

$$f_1(x) + f_2(x) - \frac{1}{4}x^2 - \frac{1}{2}x^2 = x$$

$$-2f_1'(x) + 2f_2'(x) + 5x^2 - x^2 = 0$$

$$f_2'(x) - f_1'(x) = -2x^2$$

$$f_2(x) - f_1(x) = -\frac{2}{3}x^3 + C$$

$$f_1(x) + f_2(x) = 3x^2 + x$$

$$f_2(x) = \frac{-\frac{2}{3}x^3 + 3x^2 + x + C}{2}$$

$$f_1(x) = \frac{\frac{2}{3}x^3 + 3x^2 + x - C}{2}$$

$$f_1(x) + f_2(x) =$$

$$u = \frac{1}{3}(x-2t)^3 - \frac{1}{3}(x+2t)^3 + \frac{3}{2}(x-2t)^2 + \frac{3}{2}(x+2t)^2 + x$$

$$11. \begin{cases} \frac{\partial^2 u}{\partial r^2} + \frac{\partial u}{r \partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0 \\ u|_{r=1} = 8\sin^4 \theta + \sin 4\theta, \quad 0 \leq \theta \leq 2\pi \end{cases}$$

$$\text{令 } u = R(r)\phi(\theta)$$

$$\frac{r^2 R''(r)}{R(r)} + \frac{r R'(r)}{R(r)} = -\frac{\phi''(\theta)}{\phi(\theta)} = \lambda$$

$$\phi''(\theta) + \lambda \phi(\theta) = 0, \quad \begin{cases} \phi(0) = \phi(2\pi) \\ \phi'(0) = \phi'(2\pi) \end{cases}$$

$$(i). \lambda < 0. \quad \phi(\theta) = A e^{\sqrt{\lambda} \theta} + B e^{-\sqrt{\lambda} \theta}$$

$$A = B = 0.$$

$$(ii). \lambda = 0. \quad \phi(\theta) = A\theta + B$$

$$A = 0 \quad \phi(\theta) = B.$$

$$(iii). \lambda > 0. \quad \phi(\theta) = A \cos \beta \theta + B \sin \beta \theta$$

$$\lambda = \beta^2 \quad \beta = n$$

$$\phi(\theta) = A_n \cos n\theta + B_n \sin n\theta$$

$$r^2 R''(r) + r R'(r) - \lambda R(r) = 0$$

$$\textcircled{1} \lambda = 0. \quad D^2 R(r) = 0.$$

$$R(r) = At + B$$

$$R(r) = A \ln r + B$$

$$\textcircled{2} \lambda > 0. \quad D^2 R(r) - n^2 R(r) = 0$$

$$R(r) = A e^{nx} + B e^{-nx}$$

$$R(r) = A r^n + B r^{-n}$$

$$\because |R(0)| < \infty. \quad B' = 0. \quad A = 0.$$

$$R(r) = A_n r^n$$

$$u(r) = A_0 + \sum_{n=1}^{\infty} (A_n \cos n\theta + B_n \sin n\theta) r^n$$

$$A_0 + \sum_{n=1}^{\infty} (A_n \cos n\theta + B_n \sin n\theta) = 8\sin^4 \theta + \sin 4\theta$$

$$= 3 - 4\cos 2\theta + \cos 4\theta + \sin 4\theta$$

$$A_0 = 3. \quad \sum_{n=1}^{\infty} A_n \cos 2\theta = -4\cos 2\theta$$

$$\sum_{n=1}^{\infty} B_n \sin 4\theta = \sin 4\theta$$

$$\sum_{n=1}^{\infty} A_n \cos 4\theta = \cos 4\theta$$

$$A_n = \frac{1}{\pi} \int_0^{2\pi} -4\sin^2 2\theta \cos n\theta d\theta$$

$$= -4$$

f

$$\sum_{n=1}^{\infty} A_n (\cos 2\theta + \cos 4\theta)$$

$$\sum_{n=1}^{\infty} A_n (\cos 4\theta + \cos 2\theta) = \cos 4\theta - 4\cos 2\theta$$

$$B_n = \frac{1}{\pi} \int_0^{2\pi} \sin^2 4\theta \sin n\theta d\theta$$

$$= 1$$

$$A_n = \frac{1}{2\pi} \int_0^{2\pi} (\cos^2 4\theta - 4\cos 2\theta) (\cos 4\theta + \cos 2\theta) d\theta$$

$$= -\frac{3}{2}$$

$$u(r) = 3 + \sum_{n=1}^{\infty} \left(-\frac{3}{2} \cos n\theta + \sin n\theta \right) r^n$$

九.

$$u(x, t) = v(x, t) + w(x)$$

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 v}{\partial t^2}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 w}{\partial x^2}$$

$$\frac{\partial^2 v}{\partial t^2} = 4 \frac{\partial^2 v}{\partial x^2} + 4 \frac{\partial^2 w}{\partial x^2}$$

$$4w'' = -\sin \frac{x}{2}$$

$$w(0) = 0. \quad w'(x) - w'(0) = 2$$

$$w(x) = \sin \frac{x}{2} + 2x$$

$$\begin{cases} \frac{\partial^2 v}{\partial t^2} = 4 \frac{\partial^2 v}{\partial x^2} \\ v|_{x=0} = 0. \quad \frac{\partial v}{\partial x}|_{x=\pi} = 0 \end{cases}$$

$$v|_{x=0} = 0. \quad \frac{\partial v}{\partial x}|_{x=\pi} = 0$$

$$v(t=0) = -\sin \frac{x}{2} \quad \frac{\partial u}{\partial t}|_{t=0} = 0.$$

$$\frac{1}{2} V(x, t) = X(x) T(t)$$

$$+ \quad \mathcal{F}[u(x, t)] = U(w, t)$$

$$X(x) T'(t) = 4 X''(x) T(t)$$

$$\frac{X''(x)}{X(x)} = \frac{T'(t)}{4T(t)} = -\lambda$$

$$X''(x) + \lambda X(x) = 0.$$

$$(i) \quad \lambda < 0. \quad X(x) = A e^{\sqrt{-\lambda}x} + B e^{-\sqrt{-\lambda}x}$$

$$A = B = 0$$

$$(ii) \quad \lambda = 0. \quad X(x) = Ax + B$$

$$A = 0, B = 0$$

$$(iii) \quad \lambda > 0. \quad X(x) = A \cos \beta x + B \sin \beta x$$

$$\lambda = \beta^2. \quad X'(x) = -A\beta \sin \beta x + B\beta \cos \beta x$$

$$A = 0.$$

$$\cos \beta \pi = 0$$

$$\beta = \left(\frac{1}{2} + n\right)$$

$$\lambda = \left(\frac{1}{2} + n\right)^2$$

$$X(x) = C \sin\left(\left(\frac{1}{2} + n\right)x\right)$$

$$T''(t) + 4\lambda T(t) = 0.$$

$$T(t) = A_n \cos(1+2n)t + B_n \sin(1+2n)t$$

$$U(x, t) = \sum_{n=1}^{\infty} [A_n \cos(1+2n)t + B_n \sin(1+2n)t] \sin\left(\left(\frac{1}{2} + n\right)x\right)$$

$$\sum_{n=1}^{\infty} A_n \sin\left(\left(\frac{1}{2} + n\right)x\right) = 2x$$

$$\sum_{n=1}^{\infty} (1+2n) B_n \sin(1+2n)t = 0.$$

$$B_n = 0. \quad A_n = \frac{2}{\pi} \int_0^{\pi} 2x \sin\left(\left(\frac{1}{2} + n\right)x\right) dx$$

$$= \frac{16}{\pi(1+2n)^2} \sin\left(\left(\frac{1}{2} + n\right)\pi\right)$$

$$U(x, t) = \sum_{n=1}^{\infty} \frac{16}{\pi(1+2n)^2} \sin\left(\left(\frac{1}{2} + n\right)\pi\right) \cos(1+2n)t \sin\left(\left(\frac{1}{2} + n\right)x\right) + \sin \frac{x}{2} + 2x$$